

Practice Problem Set for the Final Exam

ECON 320 – Introduction to Mathematical Economics

Fall 2025

Problem 1 (Functions and Graphs)

Consider the function

$$f(x) = \frac{3x^2 - 4x + 5}{x + 2}.$$

1. Determine the domain of $f(x)$.
2. Compute the horizontal and vertical asymptotes, if any.
3. Sketch the basic shape of the function and identify any intercepts.

Problem 2 (Linear Models and Elasticity)

A market demand function is given by

$$Q_d = 120 - 2P, \quad Q_s = -20 + 3P.$$

1. Find the equilibrium price and quantity.
2. Compute the price elasticity of demand at the equilibrium.
3. Determine the price elasticity of supply at the equilibrium.

Problem 3 (Differentiation)

Let

$$y = (3x^2 + 5x)e^{0.2x}.$$

1. Compute y' using appropriate differentiation rules.
2. Compute the second derivative y'' .

Problem 4 (Single-Variable Optimization)

A firm has the cost function

$$C(Q) = 20Q + 5Q^2,$$

and the inverse demand function

$$P(Q) = 100 - Q.$$

1. Write the profit function $\pi(Q)$.
2. Find the first-order condition for profit maximization.
3. Determine the critical point(s).
4. State whether each critical point is a maximum or minimum (you do not need to justify).

Problem 5 (Exponential and Logarithmic Functions)

Suppose a country's GDP evolves according to

$$Y(t) = 500e^{0.04t}.$$

1. Compute the instantaneous growth rate of $Y(t)$.
2. Find $Y(20)$.
3. Calculate the elasticity of Y with respect to t .

Problem 6 (Multivariable Optimization)

Let

$$f(x, y) = 20x + 10y - x^2 - y^2 - 2xy.$$

1. Compute the first-order partial derivatives f_x and f_y .
2. Find all stationary points.
3. Compute the Hessian matrix.
4. Evaluate the determinant of the Hessian at the stationary point(s).
5. Classify each stationary point as a maximum, minimum, or saddle point.

Problem 7 (Constrained Optimization)

A consumer maximizes the utility function

$$U(x, y) = x^{1/3}y^{2/3}$$

subject to the budget constraint

$$4x + 2y = 120.$$

1. Form the Lagrangian for this problem.
2. Derive the first-order conditions.
3. Solve for the optimal bundle (x^*, y^*) .
4. Write the bordered Hessian matrix for this problem (no need to evaluate it).

Problem 8 (Envelope Theorem)

Consider the maximization problem

$$\max_x f(x, b) = bx - x^2,$$

where b is a parameter, and there are no constraints on x .

1. Find the first-order condition for interior solutions.
2. Derive the optimal $x^*(b)$.
3. Compute the value function $v(b) = f(x^*(b), b)$.
4. Use the envelope theorem to compute $v'(b)$ without differentiating $x^*(b)$.

Problem 9 (Integration)

Evaluate each of the following integrals:

1. $\int (6x^3 - 4x + 5) dx$
2. $\int_0^2 e^{3t} dt$
3. $\int \frac{2x + 1}{x^2 + x} dx$

Problem 10 (Economic Dynamics)

Consider the differential equation

$$\frac{dy}{dt} + 3y = 12.$$

1. Solve the differential equation for the general solution $y(t)$.
2. Suppose $y(0) = 0$. Find the particular solution.
3. Interpret the long-run behavior of $y(t)$.